

Guide to a Limping Child in Children's Emergency

DEFINITION:

A limp is defined as an abnormal gait, and may be caused by pain, deformity or weakness.

This guide focusses on the approach to a painful limp, with a flow chart for the child with painful limp who presents to the Children's Emergency (CE).

CONTENTS

1. [Causes of painful limp](#)
2. [FLOW CHART for A CHILD WITH A LIMP IN CE](#)
3. [History](#)
4. [Physical Examination](#)
5. [Investigations](#)
6. Acute joint pain (< 1 week)
 - A) [With fever](#)
 - B) [Without fever](#)
 - C) [Additional Management and Disposition](#)
7. [Subacute / Chronic joint pain](#)

1) CAUSES OF PAINFUL LIMP

The differentials are wide and the conditions not to be missed are: **infection, malignancy, non-accidental injury.**

First, establish that the limp is due to pain, not (neuromuscular) weakness or deformity.

Then, try to localize the area of pain:

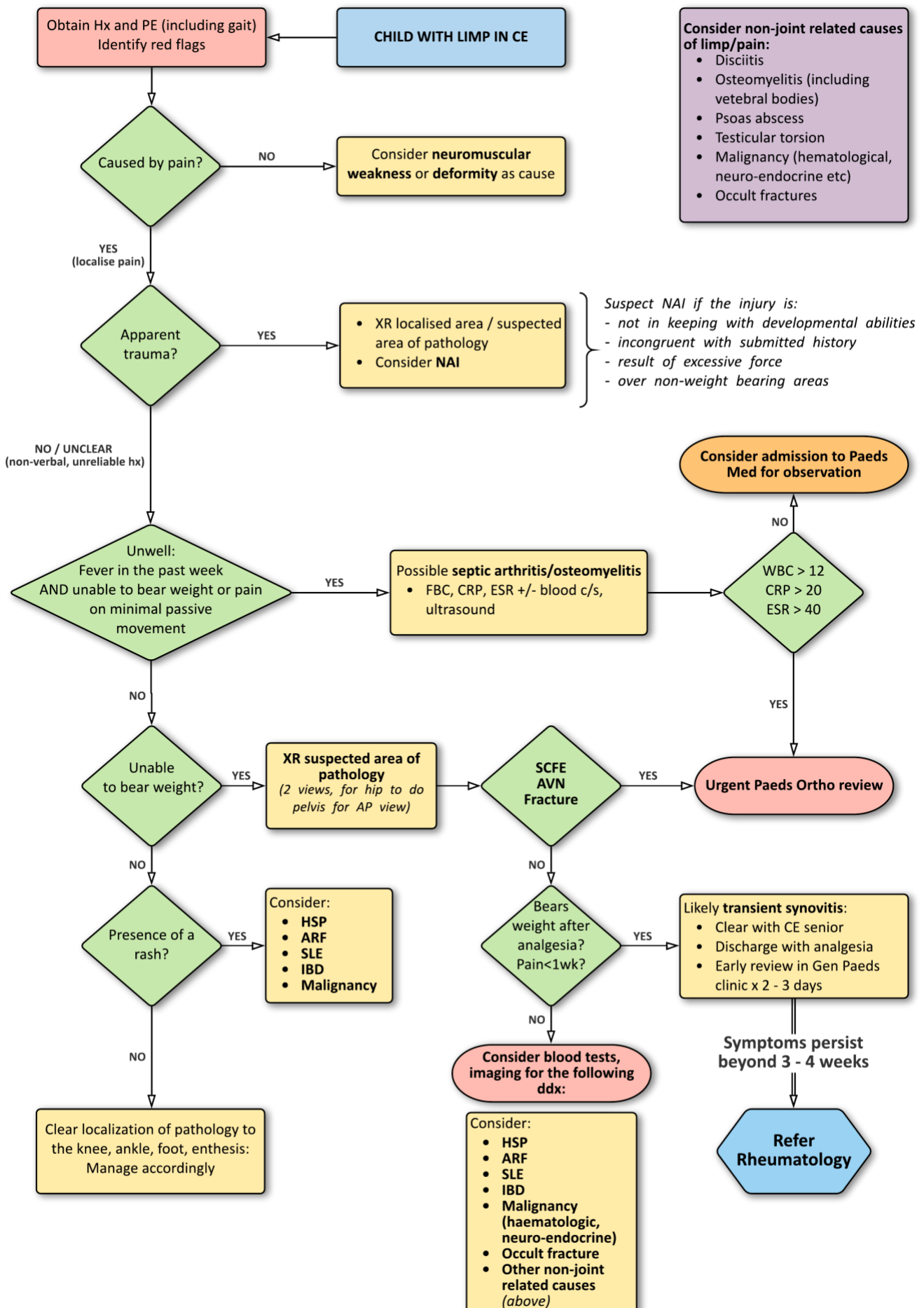
- back, pelvis, hip, thigh, knee, calf, ankle, foot
- skin, soft tissue, muscle, bone or joint

Table 1: Causes of PAINFUL limp in paediatric patients

All ages		
- Trauma (fracture / dislocation, sprain / strain, haemarthrosis, soft tissue contusions) - including non-accidental injury (NAI) - Infection (septic arthritis, osteomyelitis, discitis, psoas abscess) - Malignancy (haematological and solid tumours) - Post-infectious (reactive arthritis / transient synovitis, acute rheumatic fever (ARF)) - Juvenile Idiopathic Arthritis (JIA) - Mechanical or overuse (apophysitis, osteochondroses) - Referred pain (e.g. appendicitis, testicular / ovarian disorders, etc) - Non-malignant hematologic disease e.g. hemophilia, sickle cell disease - Metabolic disease e.g. rickets - Psychosomatic		
Common causes by age		
Toddler (1 – 3 years)	Child (4 – 10 years)	Adolescent (11 – 16 years)
- Foreign body or minor laceration in foot - Septic arthritis, osteomyelitis - Transient synovitis - Toddler's fracture - NAI - Inflammatory (e.g. HSP, JIA) - Haemophilia hemarthrosis	- Transient synovitis - Septic arthritis, osteomyelitis - Perthes Disease - Inflammatory (e.g. HSP, JIA, enthesitis, part of systemic illness such as IBD, SLE, etc.) - Haemophilia haemarthrosis	- SCFE - Mechanical: osteochondritis dissecans, apophysitis (e.g. Osgood-Schlatter Disease, Sever's Disease) - Inflammatory (e.g. JIA, enthesitis, etc.) - JIA - Septic arthritis, osteomyelitis - Malignancies esp osteosarcoma - Psychosomatic - Sexually transmitted illnesses (STI) with arthralgias and arthritis
<i>The young child... ...should not be diagnosed as having psychosomatic limping ...with JIA rarely complains of pain</i>		

HSP Henoch Schönlein Purpura, IBD inflammatory bowel disease, SLE Systemic Lupus Erythematosus, SCFE Slipped Capital Femoral Epiphysis

2) FLOW CHART FOR A CHILD WITH A LIMP IN CE



3) HISTORY

1. **Age** of patient
2. **Localization**
 - a. Skin and soft tissue
 - b. Muscle
 - c. Bone
 - d. Joint (*beware of knee pain that is actually referred hip pain*)
 - e. Referred pain – lower abdominal structures (e.g. appendicitis, psoas abscess, genitalia issues (e.g. testicular disorders))
3. **Duration** of symptoms
 - a. Acute: < 1 week
 - b. Subacute: 1 – 4 weeks
 - c. Chronic: 4 – 6 weeks
4. **Character** of pain
 - a. Nocturnal pain (infection or malignancy)
 - b. Inflammatory versus mechanical pain (Table 2)
 - c. Migratory (ARF)
 - d. Severity – functional history

Table 2: Differentiating inflammatory and mechanical pain

Factors	Inflammatory arthritis	Mechanical pain
Pain - time of day - relation to activity	Better with activity; Worst in morning , better by evening	Worse with activity; Worst in evening , improves with rest. Usually no or minimal pain in morning
Early morning stiffness (EMS)	>30min	<15min
Swelling	Yes	Maybe
Warmth	Sometimes	No
Limitation of movement	Yes ("limping", "can't bend")	No, unless anatomic derangement

5. Significant **trauma** (*Beware of minor trauma as a red herring.*) (*This history may not be available in a young/non-verbal child.*)
6. **Associated symptoms**

Symptom	Differential
Fever	Septic arthritis, osteomyelitis, malignancy
Constitutional symptoms – anorexia, weight loss	Malignancy, arthritis as part of a systemic inflammatory condition
Rash	HSP, ARF, SLE, IBD, malignancy

7. **Risk factors** for different etiologies

Risk factor	Differential
Open wounds, poor skin condition (e.g. poorly controlled eczema)	Septic arthritis, osteomyelitis
Recent history of infection	Transient synovitis
Activity in sports	Stress fractures, over use syndromes, osteochondritis
Family history of autoimmune disorders	JIA, SLE, IBD, sarcoid, etc.
Complex social setup	Non-accidental injury
Psychosocial issues	Psychosomatic pain
Sexual activity, multiple partners/unprotected sex, previous history of STI	STI with arthralgia/arthritis; e.g. gonococcal, reactive arthritis

RED FLAGS

Sick or toxic child (think of septic arthritis, osteomyelitis, malignancy)
 Complete refusal to weight bear
 Severe limitation to range of movement

4) PHYSICAL EXAMINATION

Important features to look for in physical examination:

1. **General:** well VS unwell/toxic
2. **Gait-** the limb that is taking weight for the shortened time harbors the pathology
 - a. Walk
 - b. Run (exacerbates subtle abnormalities in gait)
 - c. Squat (quick screen for hip, knee and proximal muscle power)If child is unable to cooperate, parents may be able to imitate the gait, or provide videos.
3. **Back or leg tenderness** – Examine the whole leg (not just “affected joint”), skin, soft tissue, muscle, bone
4. **Limb**-length discrepancy
5. **Detailed joint examination** – swelling, erythema, warmth, tenderness, limitation in range of movement
 - a. Abduction and internal rotation usually affected in hip pathologies
 - b. Don't forget FABER test for sacro-iliac joint pathology

Note: If *joint* pathology is found to be the cause of limp, do a **pGALS** to quickly screen for other joint/spine pathology.

6. **Neurology**
7. **Skin/mucosa:** bruising, petechiae, pallor, rash (vasculitic vs non-vasculitic), ulcers
8. **Abdomen:** screen for hepatosplenomegaly and tenderness. Do not forget to screen for testicular pathologies.
9. **Lymphadenopathy**, especially in groin

5) INVESTIGATIONS

Investigations such as blood tests and imaging may often be normal in many of the causes and should be guided by clinical suspicion based on history and clinical examination.

FBC, ESR, CRP	Useful in the febrile patient, or those in whom infection (esp septic arthritis) is being considered. Consider these even in the afebrile child with normal radiography.
Blood culture	For those in whom septic arthritis and osteomyelitis are being considered.
Radiographs	To look for fracture, Perthes disease, SCFE, bone lesions (malignancy, osteochondroses, benign tumours), congenital abnormalities. Normal in early osteomyelitis.
Ultrasonography	Useful for detecting small joint effusions of the hip. Consider when Xrays are normal but the suspicion of septic arthritis remains high.
MRI	Rarely done in the emergency setting.
Bone scans	Almost never done in the emergency setting.

6) ACUTE JOINT PAIN

A) With Fever ($T \geq 38$ deg C)

Differentials:

1. **Infection**** (*usually single joint*)
 - a. Septic arthritis
 - b. Osteomyelitis
2. Other systemic infections:
 - a. Viral: Influenza and Dengue / Chikungunya (*arthralgia*), Infectious Mononucleosis (*can have arthritis*)
 - b. Bacteria: Mycoplasma (*can have arthritis*)
3. Systemic JIA (SJIA) (*usually > 1 joint, look for associated features of rash, hepatosplenomegaly and lymphadenopathy*)
4. Malignancy (*usually subacute, nocturnal pain prominent*)
 - a. Acute leukaemia
 - b. Neuroblastoma
 - c. Bone metastasis
5. Acute Rheumatic Fever (*refer to Jones criteria for diagnosis*)
6. Gout (*older age group with underlying renal disease, hyperuricaemia or metabolic syndrome*)

Investigations:

1. FBC, CRP, ESR
 2. XR of joint/limb (2 views; for hip XR, replace anterior view with XR pelvis AP instead)
- ± *Peripheral blood film, liver function test, renal panel 2 if a haematological malignancy is suspected*
± *Ferritin, liver function test if SJIA is suspected*
± *Renal panel 2 if gout is suspected*

**Have a low index of suspicion for septic arthritis. Admit and refer the patient to Paediatric Orthopaedics if he/she has any of the following:

1. Fever + unable to weight bear/severe restriction of movement
2. Satisfies ≥ 2 Kocher criteria (table 3):
 - Fever > 38.5 deg C
 - Inability to weight bear
 - ESR > 40 and/or CRP > 20 mg/L
 - WBC $> 12 \times 10^9/L$

Points	Likelihood of Septic Arthritis
0	0.20%
1	3%
2	40%
3	93%
4	99%

Table 3: Likelihood of septic arthritis based on Kocher criteria.

B) Without Fever

Differentials:

1. Trauma – fracture, sprain, strain
 - a. NAI needs to be excluded in children who are non-communicative or have special needs
2. Transient synovitis / reactive arthritis
3. Henoch Schonlein Purpura (HSP)
4. Early presentation of septic arthritis / osteomyelitis (*consider if unable to weight bear / sig reduced ROM*)
5. Haemarthrosis
 - a. Systemic disease: coagulopathy e.g. haemophilia, severe von Willebrand disease
 - b. Local disease: pigmented villonodular synovitis, synovial haemangioma
6. Early presentation of conditions that cause chronic joint pain
 - a. Inflammatory: JIA (except SJIA), IBD, SLE, etc.
 - b. Tumour – benign VS malignant, bony VS non-bony
 - c. Orthopaedic conditions e.g. Perthes disease, SCFE, Apophysitis (e.g. Osgood Schlatter and Sever's Disease)
7. Gout (*older age group with underlying renal disease, hyperuricaemia or metabolic syndrome*)

Suspect NAI if the injury is:

- Not in keeping with developmental abilities of the child
- A result of excessive force
- Incongruent with submitted history
- Over non-weight bearing areas

Investigations:

- **XR of joint (2 views; for hip XR, replace anterior view with XR pelvis AP instead)**
- *If pain is not well localized to a joint, then XR of the limb ± joint above and below can be considered*
- ± FBC/ESR/CRP if septic arthritis is a consideration
- ± Other investigations depending on clinical suspicion

C) Additional Management and Disposition *(list is not exhaustive)*

Working Diagnosis	Additional management	†Disposition
Septic arthritis / osteomyelitis	Refer Orthopaedics → <i>urgent joint aspiration ± arthrotomy and washout may be required in septic arthritis (Keep patient nil by mouth)</i> Consider ultrasound Blood culture and early IV antibiotics after discussion with Orthopaedics	Admit to Orthopaedics
Trauma – fracture, sprain, strain	Manage as per clinical condition Refer to Orthopaedics if necessary Analgesia	Admit or discharge with Orthopaedics (TCU as per clinical condition)
Transient synovitis	†NSAIDs at anti-inflammatory dose	Admit for observation if unable to weight bear Discharge with red flag advice for septic arthritis and TCU General Paediatrics in 2-3 days
SJIA	Additional investigations: LFT, ferritin Blood culture and early IV antibiotics if bacterial sepsis is a differential	Admit to Paediatric Medicine
JIA (except SJIA) and other inflammatory arthropathies	†NSAIDs at anti-inflammatory dose	TCU Paediatric Rheumatology within 2 weeks
HSP	Urine dipstick and urine microscopy Blood pressure †NSAIDs at anti-inflammatory dose	Admit to Paediatric Medicine if: • Inability to weight bear despite adequate analgesia • Abdominal pain • Per rectal bleeding • Hypertension • Proteinuria Discharge with red flag advice and early Paediatric Medicine TCU if red flags absent
Acute Rheumatic Fever	Additional investigations: ASOT, Electrocardiogram (ECG), 2D-Echocardiogram (2DE) (non-urgent)	Admit to Paediatric Medicine
Gout	Additional investigation: RP2 PO colchicine – dose to be discussed with primary nephrologist / rheumatologist-on-call †NSAIDs at anti-inflammatory dose (<u>only if no underlying chronic kidney disease</u>)	Admit to Paediatric Medicine if unable to weight bear despite adequate analgesia. Discharge with red flag advice for septic arthritis and TCU Paediatric Nephrology (if known) or TCU Paediatric Rheumatology early (3 – 5 days)
Malignancy	Additional investigations for suspected haematological malignancy: Peripheral blood film (PBF), RP2, LFT	Admit to Paediatric Medicine Co-manage with Orthopaedics if bony malignancy
Perthes disease, SCFE	Refer Orthopaedics SCFE usually requires emergency pinning and reduction	Dependent on Orthopaedics plans
Apophysitis	Analgesia, avoidance of physical activity that can exacerbate condition	TCU Orthopaedics x 1 – 2 weeks

†All patients should be **cleared with the CE senior on shift prior to discharge**.

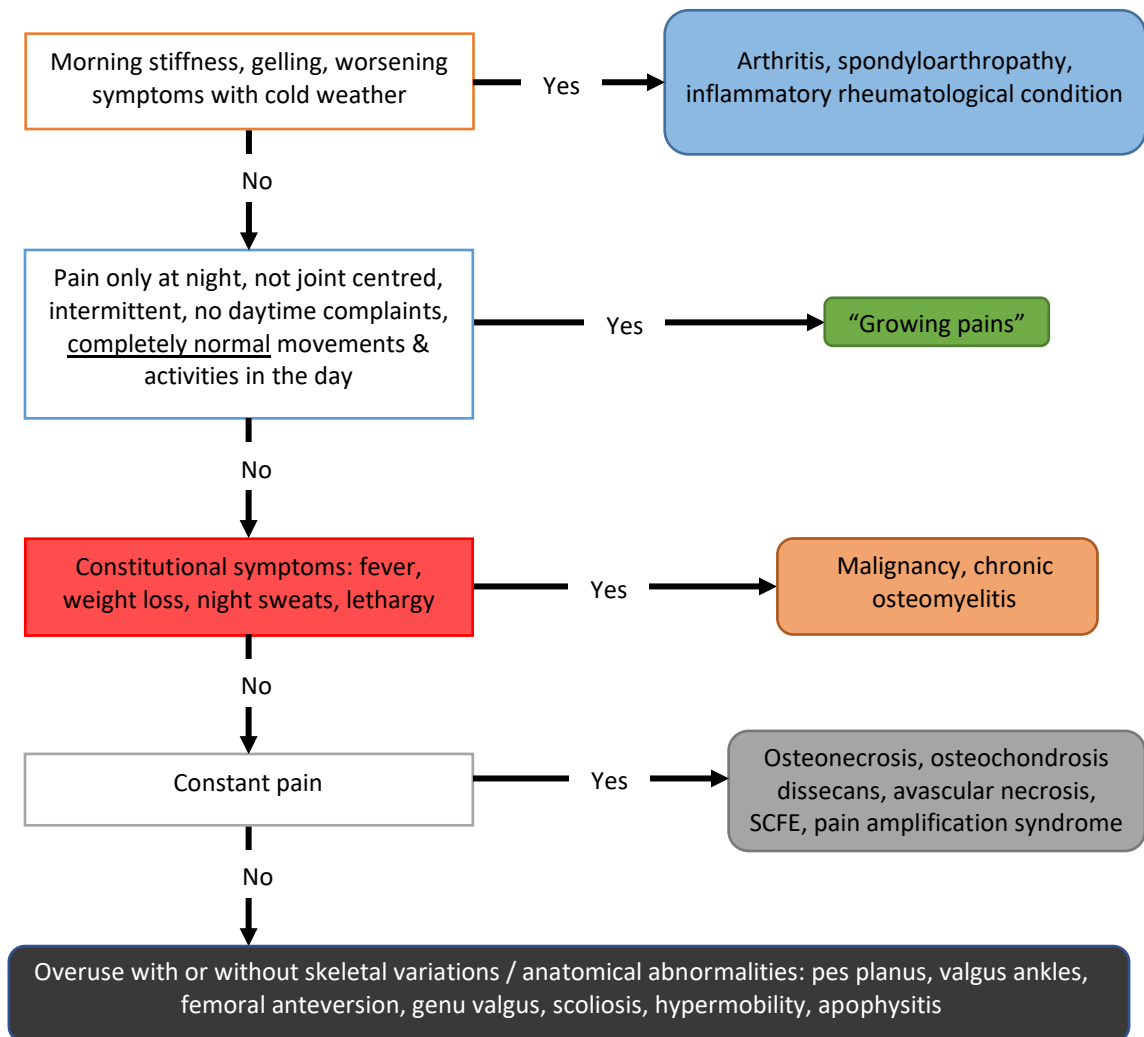
†Anti-inflammatory dose of NSAIDs:

- PO ibuprofen 10 mg/kg TDS (with/after food): *suspension 20 mg/mL, 200 mg tablet*
- PO naproxen 10 mg/kg BD (with/after food): *275 mg tablet*

Consider prescribing gastric protection medications (H2-blockers or proton pump inhibitors) if patient has previous history of gastritis.

7) SUBACUTE/CHRONIC JOINT PAIN

General approach (adapted from *Pediatr Clin N Am* 59 (2012) 471 – 492):



Investigations

- Blood tests and imaging should be tailored based on the differentials in mind

Management and Disposition

- Patient should be admitted for further work-up if red flags are present in history and physical examination
- Same-day Orthopaedics referral should be made for Orthopaedic conditions that require urgent care e.g. SCFE, avascular necrosis, etc.
- Prior to discharge, all patients will need to be cleared with the CE senior on shift
- Anti-inflammatory doses of NSAIDs should be prescribed on discharge if an inflammatory arthritis is suspected (*see previous page for dose*)
- A follow-up with Paediatric Rheumatology should be arranged in 1 – 2 weeks for patients deemed fit for discharge, unless an Orthopaedic condition is clearly diagnosed.

Source: Drs Pauline Chan, Ivy Ang, Seo Woon Li, Ian Tan, Elizabeth Ang (October 2020)

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